Colegio de Montalban

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Institute of Computing

“Learning Management System for Institute of Computing Studies at Colegio de Montalban”

An Undergraduate Thesis Presented to the

Institute of Computing Studies

Colegio de Montalban

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In partial fulfillment of the

requirements for the degree

Bachelor of Science in

Information Technology

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May 2026

CHAPTER 1: INTRODUCTION

1.1 BACKGROUND OF THE STUDY

International university studies show that a Learning Management System (LMS) can provide support for planning, monitoring, and reviewing learning alongside access to course materials. Du et al. (2025) examined 69 management undergraduates taking mathematics at a large public university in China, whose name was not disclosed. Students using its self-regulated learning support reported stronger self-regulation; an examination advantage appeared on the first test but not the second. Similarly, Çakiroğlu et al. (2024) studied a third-year Operating Systems and Applications course at a public university in Turkey. They found significant associations between assessment interactions and goal setting, self-evaluation, and help-seeking. The Chinese study evaluated an intervention, whereas the Turkish study examined interaction patterns. Together, they support considering how academic information and learning support work within the same environment, without assuming that system access alone improves learning.

Research in computing education also indicates that engagement mechanisms need to serve a defined instructional purpose. Ahmed et al. (2025) evaluated a gamified, mobile, cloud-based LMS with 60 second-year students taking Object-Oriented Programming at Mansoura University in Egypt. Their experimental group obtained higher post-test achievement and engagement scores than the comparison group. In a broader systematic review, Jaramillo-Mediavilla et al. (2024) reported favorable motivation and academic-performance findings in parts of the reviewed gamification literature. The Mansoura study provides evidence from a specific computing course, while the review draws together findings from multiple educational settings. These studies justify examining participation support in an institutional LMS, but their educational results cannot be presumed for a newly developed system at Colegio de Montalban.

In the Philippines, studies show that the organization of an LMS affects how students and instructors experience academic tasks. Llamas et al. (2025) examined the Moodle-based LMS of Mindanao State University–Iligan Institute of Technology in Iligan City through discussions and task

interaction involving six academics and six fourth-year students. They identified improvement areas in usability, communication, course building, monitoring and assessment, content management, and technical support. Laspiñas and Almerez (2024) surveyed 33 business graduate students at an unnamed university in Cebu City and reported generally neutral task-related and experience-related evaluations. The MSU–IIT study identified workflow concerns, while the Cebu study described users' overall experience. Both support examining clarity and usability through actual academic tasks instead of treating the availability of a platform as sufficient evidence of its suitability.

Within Colegio de Montalban, the present study focuses on the Institute of Computing Studies (ICS), including the academic workflows of its students and instructors and the administrative functions supporting them. Computing education requires coordination of learning resources, coursework, programming activities, assessment records, and faculty responsibilities. The research group's initial project description identifies the use of separate tools for academic work and personal organization as a concern to be addressed. This description serves as the starting problem context, while the extent of fragmentation and users' specific difficulties remain matters for local requirements gathering.

The central development problem is how to organize these related activities within a shared environment while maintaining clear responsibilities and appropriate access to information. Students need understandable access to their coursework and academic records; instructors need manageable teaching workflows; academic supervisors need suitable course-level information; and administrators need controlled system-management functions. A connected design must also account for schedules, changes in class occurrences, and the privacy of personal academic information. Combining these responsibilities requires more than placing several tools on the same screen.

In response, this study proposes the CdM Learning Management System (CdM LMS), a multi-platform environment tailored to ICS. Its interconnected applications will provide complementary access to shared information, with the Web Portal serving as the principal academic environment, the Mobile Application supporting selected academic and productivity activities, and

the Desktop Application serving as the Administrator Console. The study will develop and evaluate the system within the ICS setting. Its contribution will be evidence about the suitability and quality of this institutional implementation, rather than an assumed improvement in students' grades or long-term learning outcomes.

1.2 STATEMENT OF THE PROBLEM

This study seeks to determine the level of software quality of the LMS for ICS at Colegio de Montalban in supporting coordinated academic resource and grade management, learning support, academic coordination, and system administration. The evaluation will follow six selected characteristics of the ISO/IEC 25010:2023 Software Product Quality Model (International Organization for Standardization [ISO], 2023).

Specifically, the study seeks to answer the following questions:

1. What is the level of the LMS in supporting academic resource and grade-management workflows in terms of functional suitability, as measured by:

- **Functional completeness** in supporting course materials, assignments, dynamic grading, CodeLab activities, Git-integrated submissions, and the defined academic coordination functions;

- **Functional correctness** of weighted grade computations, EXP and level calculations, descriptive analytics, and schedule revisions and exceptions;

- **Functional appropriateness** in supporting connected academic workflows for BSIT and BSCpE users within their assigned roles and platform responsibilities?

2. What is the level of the system in providing understandable and engaging task interaction in terms of interaction capability, as reflected in:

- **Appropriateness recognizability** of course, grading, productivity, gamification, and academic coordination functions;

• **Learnability** of permitted tasks in the Web Portal, selected Mobile Application functions, and administrator-only Desktop Application;

• **User engagement** supported by EXP, levels, achievements, leaderboards, study tools, and the moderated Knowledge Exchange environment?

3. What is the level of the system in protecting academic information and controlling access in terms of security, as demonstrated by:

• **Confidentiality** of student grades, personal notes, academic records, and restricted course information;

• **Integrity** of grades, submissions, schedule records, and events during authorized updates and concurrent actions;

• **Accountability** through activity and audit records for designated administrative actions, academic coordination changes, and Knowledge Exchange moderation?

4. What is the level of the system in processing academic information and maintaining applicable shared records in terms of performance efficiency, as indicated by:

• **Time behavior** in computing grades, generating descriptive analytics and calendar views, retrieving course resources, and displaying relevant in-LMS updates;

• **Resource utilization** during the processing and retrieval of shared information by the authorized Web, Mobile, and Desktop components under defined workloads?

5. What is the level of the system in providing stable access to permitted academic and study functions in terms of reliability, as evaluated through:

• **Availability** of the Web Portal, supported mobile study tools, and Administrator Console during the defined evaluation period;

• **Fault tolerance** when processing interrupted requests, concurrent grade updates, or simultaneous Knowledge Exchange activity, including the handling of failed operations without incomplete or duplicate records?

6. What is the level of the system in supporting consistent information exchange and operation alongside other software in terms of compatibility, as indicated by:

• **Interoperability** among the Web Portal, Mobile Application, and Administrator Console through their applicable shared records and authorized backend workflows;

• **Co-existence** of the Administrator Console with other software in the supported administrator workstation environment without adversely affecting their operation?

1.3 Statement of the Objectives

1.3.1 General Objective

To develop and evaluate a Learning Management System for the Institute of Computing Studies at Colegio de Montalban that coordinates academic management, learning support, academic oversight, and system administration through shared information and role-appropriate access.

1.3.2 Specific Objectives

The study specifically aims to:

1. Establish shared backend services and a centralized PostgreSQL database for the principal Web Portal, complementary Mobile Application, and administrator-only Desktop Application, with credential-based authentication, verified and approved accounts, and server-enforced access for Student, Instructor, Super User, and Administrator roles.

2. Develop course and section management that allows authorized instructors to organize materials, annotations, assignments, announcements, and deadlines, and allows enrolled students to access resources and submit permitted work under the applicable validation rules.

3. Implement a Dynamic Gradebook with configurable assessment components and weights, score recording, automatic weighted-grade computations, personal student grade views, instructor performance summaries, and CSV grade export; verify computed values and charts against documented test records.

4. Provide CodeLab for in-browser programming execution and assessment against predefined test cases, and a separate Git-integrated workflow for repository-based coursework submission and instructor review; verify the supported submission and access rules for each workflow.

5. Develop notes, task lists, flashcards, review and mastery records, focus timers, and study-session tracking, together with rule-based EXP, levels, achievements, badges, streaks, and department leaderboards; verify progression calculations and the separation of EXP from course grades and forum reputation.

6. Implement Knowledge Exchange for academic questions, answers, and resource sharing, with authorized instructor moderation, Faculty-Verified Answers, contribution-based reputation, and records of designated moderation actions.

7. Develop the proposed faculty directory and academic operations hub, distinguishing program affiliations and Dean or Program Head designations from primary account roles, and providing restricted course supervision that excludes student identities, individual grades, submissions, assessment attempts, and answers.

8. Implement Asia/Manila-based weekly course schedules with effective-dated revision history and future occurrence cancellation, rescheduling, and restoration; provide authorized calendar views, audience-targeted saved events, coursework deadlines, and private note–event snapshot conversion, and verify their visibility and preservation rules.

9. Provide the Administrator Console with account approval, status and role management, permitted configuration, activity and audit records, reporting, and the backup and recovery functions

supported by the evaluated release; verify that administrative permissions remain separate from academic oversight.

10. Evaluate the developed system in terms of functional suitability, interaction capability, security, performance efficiency, reliability, and compatibility using role-appropriate tasks, feedback from eligible ICS students and instructors, and technical checks by qualified IT experts, and document defects and limitations identified in the evaluated version.

1.4 CONCEPTUAL FRAMEWORK

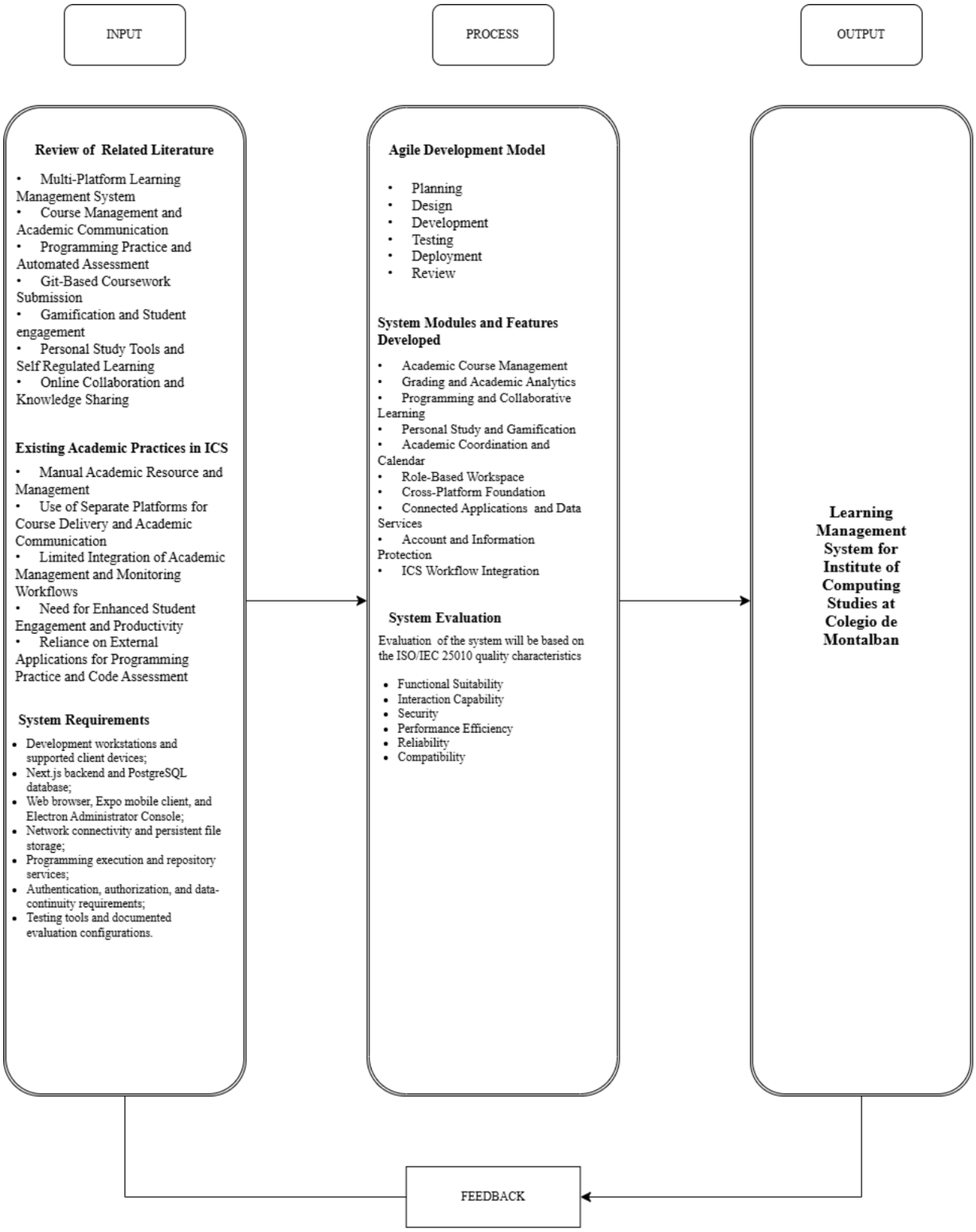


Figure 1

A Conceptual Framework Illustrating the Development of the

The conceptual framework of the study, *"Learning Management System for Institute of Computing Studies at Colegio de Montalban,"* follows the Input–Process–Output (IPO) model with a feedback loop. As illustrated in Figure 1, the framework organizes the information and resources required by the study, the development and evaluation activities, and the expected system output. Purwanto et al. (2025) examined limitations of traditional IPO representations and proposed an Input–Mediator–Output–Input (IMOI) adaptation that treats outputs and stakeholder feedback as inputs to subsequent iterations. Their discussion informs the feedback mechanism incorporated into the present study's IPO framework. The researchers will retain the IPO structure while using feedback to refine requirements and guide succeeding Agile development cycles.

The **Input component** consists of three groups: the review of related literature, initial academic practices and concerns within the Institute of Computing Studies (ICS), and system requirements. The review of related literature covers multi-platform learning management systems, course management and academic communication, programming practice and automated assessment, Git-based coursework submission, gamification and student engagement, personal study tools and self-regulated learning, and online collaboration and knowledge sharing. These topics provide the academic foundation for identifying relevant functions, understanding their limitations, and determining how they may support the intended users.

The researchers will also examine initial concerns identified within ICS, including manual academic-resource management, the use of separate platforms for course delivery and communication, limited integration of academic management and monitoring, and reliance on external applications for programming practice and code assessment. Requirements gathering will determine the extent of these conditions and identify the needs of ICS students and instructors. These initial concerns will guide investigation rather than be treated as established findings before supporting local evidence is collected.

System requirements include the technical resources and operating conditions needed to develop and evaluate CdM LMS. These comprise development workstations and supported client devices, the Next.js backend and centralized PostgreSQL database, compatible web browsers, the Expo mobile application, and the Electron Administrator Console. Additional requirements include network connectivity, persistent file storage, programming execution and repository services, authentication and authorization, data-continuity arrangements, testing tools, and documented evaluation configurations. The researchers will verify the selected configurations against the release used for evaluation.

The **Process component** explains how the identified inputs will be transformed into the proposed system through development, testing, and evaluation. The researchers will follow an adapted Agile approach consisting of Planning, Design, Development, Testing, Deployment, and Review. Planning will identify requirements, role responsibilities, platform coverage, and acceptance criteria. Design will establish the application structure, database relationships, workflows, and interfaces. Development will implement prioritized functions incrementally, while testing will examine expected behavior, access restrictions, record integrity, integration, and failure handling. Deployment will prepare a reviewed release for controlled evaluation, and review will identify corrections and priorities for subsequent development. Wiradarma et al. (2026) described an LMS development process involving planning, design, development, testing, release, and feedback. Their study provides an example of Agile application in LMS development, while Deployment and Review represent the stage labels adopted for the present project.

The development activities will cover academic course management, grading and descriptive academic analytics, programming and collaborative learning, personal study and gamification, academic coordination and calendar functions, role-based workspaces, complementary cross-platform functionality, connected applications and data services, account and information protection, and ICS workflow integration. These feature groups include course resources and submissions, the Dynamic Gradebook, separate CodeLab and Git coursework workflows, Knowledge Exchange,

personal productivity tools, gamified progression, faculty organization, restricted course supervision, schedules, targeted events, and private note–event snapshot conversion.

The role-based workspaces will distinguish the responsibilities of Students, Instructors, Super Users, and Administrators. Dean and Program Head will remain academic designations rather than additional primary account roles. The Web Portal will provide the principal academic and coordination environment, the Mobile Application will support selected academic and productivity functions, and the native Desktop Application will serve as the Administrator Console. Feature availability will follow the permissions and platform coverage of the evaluated release rather than imply identical functionality across all applications.

Developed increments will undergo functional and technical testing throughout the Agile cycles. The designated evaluation release will subsequently be assessed using six selected characteristics of ISO/IEC 25010:2023: **Functional Suitability, Interaction Capability, Security, Performance Efficiency, Reliability, and Compatibility** (International Organization for Standardization, 2023). Eligible ICS students and instructors will evaluate functions they have used through role-appropriate tasks, while qualified IT experts will examine relevant technical and administrative behavior. Questionnaire ratings will be considered alongside technical checks of calculations, permissions, record consistency, response times, failure handling, and information exchange. Abid et al. (2024) supports the importance of systematic LMS evaluation through a framework based on features and functions; however, the selected quality characteristics for this study will be derived from ISO/IEC 25010:2023.

The **Output component** represents the expected developed and evaluated CdM LMS for ICS at Colegio de Montalban. The system is intended to organize academic management, computing coursework, personal study, collaborative learning, academic oversight, and administrative functions around shared information and authorized access. Although the application supports configuration for other institutes, the research evaluation and interpretation of findings will remain within the ICS

setting. The output will reflect the functions demonstrated in the evaluated release and will not, by itself, establish improvements in grades, motivation, or long-term learning outcomes.

The **Feedback component** represents findings obtained from testing, evaluation, and review, including identified defects, task difficulties, technical observations, and relevant comments from users and evaluators. As shown by the return path in Figure 1, these findings will inform the Input component by refining requirements, operating conditions, and development priorities. The revised inputs will then guide subsequent Agile development, testing, and evaluation cycles. This feedback loop allows the researchers to correct verified problems and refine the system while documenting unresolved limitations.



1.5 OPERATIONAL FRAMEWORK

Figure 2

An Operational Framework Illustrating the Development of the Learning Management System Using the Agile Model

The operational framework of the study will follow an adapted Agile development approach consisting of Planning, Design, Development, Testing, Deployment, and Review. These phases will operate iteratively, allowing requirements, interfaces, and system functions to be refined through testing and feedback. Each cycle will address prioritized requirements and produce a system increment for examination. Wiradarma et al. (2026) described an LMS development process involving planning, design, development, testing, release, and feedback. The present study will adapt this approach to the development and evaluation of CdM LMS within the Institute of Computing Studies (ICS) at Colegio de Montalban.

Planning Phase. The researchers will gather and analyze the academic and operational requirements of ICS students, instructors, and authorized academic and administrative users. Llamas et al. (2025) identified LMS user-experience concerns involving usability, communication, course organization, assessment, content management, and technical support. Guico (2026) identified institutional concerns involving preparation time, access to ICT resources, and continuing training. These findings will inform the areas examined during local requirements gathering rather than establish the conditions within ICS. The researchers will define role responsibilities, platform coverage, and access restrictions, then organize requirements into a prioritized backlog with acceptance criteria.

Design Phase. The researchers will translate the identified requirements into the system architecture, database relationships, workflows, and interfaces. Flowcharts, an Entity Relationship Diagram, and Data Flow Diagrams will illustrate applicable processes and information movement. Laspiñas and Almerez (2024) examined task-related and experience-related qualities of an institutional LMS, supporting attention to users' interaction with the platform. Accordingly, the design will emphasize understandable navigation, task labels, and feedback while protecting personal academic information and separating academic supervision from teaching and system administration.

Development Phase. The researchers will implement selected requirements incrementally using the Next.js backend, centralized PostgreSQL database, and applicable client technologies. Development will cover academic management, grading, programming coursework, personal study, gamification, collaborative learning, academic coordination, and administrative functions. The Web Portal, Mobile Application, and Desktop Administrator Console will provide complementary functions according to their assigned responsibilities. Each increment will be connected to its related records and checked against its acceptance criteria before broader testing.

Testing Phase. The researchers will examine functional correctness, authorization, record consistency, interface behavior, integration, and failure handling using controlled accounts and test data. Checks will include grade computations, programming assessments, schedule revisions, event visibility, and private note–event conversion where enabled. The designated evaluation release will be assessed through role-appropriate tasks and technical checks aligned with six selected characteristics from the ISO/IEC 25010:2023 product-quality model: Functional Suitability, Interaction Capability, Security, Performance Efficiency, Reliability, and Compatibility (International Organization for Standardization [ISO], 2023). Identified defects will be corrected and retested.

Deployment Phase. A reviewed system release will be configured in a controlled environment for evaluation. The researchers will verify application startup, backend connectivity, account access, supported client configurations, and available data-continuity procedures. Eligible participants will receive guidance before performing assigned tasks. The release version, enabled functions, and known limitations will be recorded so that findings correspond to the system actually evaluated.

Review Phase. The researchers will examine technical results, questionnaire responses, and feedback from eligible ICS students, instructors, and qualified IT experts. Findings will identify supported behavior, unresolved defects, and areas requiring refinement. These findings will return to the development backlog to guide subsequent planning, design, implementation, and testing. This

repeated use of review and feedback is consistent with the iterative LMS development approach described by Wiradarma et al. (2026).

1.6 SCOPE AND LIMITATIONS

The study will cover the development and evaluation of CdM LMS for ICS at Colegio de Montalban. Its academic context includes BSIT and BSCPE within ICS, without restricting participation to a particular student year level. Department configuration for IBE and ITE is a technical capability of the application; those institutes are outside the research population and the intended generalization of findings.

The applications will use complementary responsibilities. The Web Portal will provide the principal environment for academic workflows and the proposed academic coordination functions. The Mobile Application will provide selected academic and productivity functions for students and instructors according to their permissions. The native Desktop Application will be dedicated to Administrator operations. The Super User will access academic oversight through the web environment and will not use the native Administrator Console. Additional Administrator access through web or mobile is not assumed.

The functional coverage includes academic and course management, grading and descriptive analytics, programming activities, collaborative learning, personal study and organization tools, gamification, academic coordination, and system administration. The feature groups and their user-facing behavior are described in Section 1.8. The expanded coordination coverage includes faculty relationships, read-only course supervision, recurring schedules with effective-dated revisions, exceptions affecting individual class occurrences, targeted events, and private note–calendar snapshot conversion.

The evaluation will use a quantitative descriptive approach and criterion-based purposive sampling involving ICS students, ICS instructors, and qualified IT experts.

A **student** respondent should:

1. Be officially enrolled in the Institute of Computing Studies of Colegio de Montalban;
2. Have prior experience using at least one Learning Management System (LMS) for academic or educational purposes;
3. Have a basic understanding of common LMS functions, such as accessing course materials, submitting assignments, viewing grades, receiving announcements, or participating in online learning activities;
4. Be willing to explore and use the implemented CdM LMS during the evaluation; and
5. Be willing to participate in the study.

An **instructor** respondent should:

1. Be an instructor affiliated with the Institute of Computing Studies of Colegio de Montalban;
2. Have prior experience using at least one Learning Management System (LMS) for teaching or academic purposes;
3. Have a basic understanding of common LMS functions, such as course management, learning material distribution, assignment management, grading, announcements, or student monitoring;
4. Be willing to explore and use the implemented CdM LMS during the evaluation; and
5. Be willing to participate in the study.

An **IT expert** respondent should:

1. Have an academic degree, professional qualification, or relevant training in Information Technology, Computer Science, Information Systems, Software Engineering, or a related field;
2. Have relevant academic or professional experience in at least one area related to software systems, such as software development, software testing or quality assurance, information systems, database systems, cybersecurity, system administration, or software engineering;

3. Have sufficient knowledge or experience to evaluate software characteristics such as functionality, usability, security, performance, reliability, or compatibility;
4. Have sufficient familiarity with software systems or Learning Management Systems to understand the functions and processes being evaluated; and
5. Be willing to examine and evaluate the implemented CdM LMS.

Limitations of the Study

The system will depend on suitable devices, network access, and the availability of its backend services. Cross-platform access does not imply identical capabilities, universal operating-system support, uninterrupted service, or full offline operation. The exact instructor functions available on mobile and the native coverage of the expanded coordination modules must be documented against the version used for evaluation. The new coordination functions are planned primarily for web access; native mobile parity is outside the stated release scope.

Academic analytics will describe available records and will not predict student outcomes. CodeLab will assess submitted programs against predefined test cases; its results will not establish complete program quality or constitute AI grading. Git repository history will support submission review, but it will not independently prove authorship or eliminate plagiarism. Recorded assessment events, such as tab switching or fullscreen exits, will be indicators for contextual review rather than conclusive evidence of cheating.

Course scheduling will use Asia/Manila as its authoritative timezone. A weekly schedule will use selected weekdays with a shared start and end time, date bounds, and related course information. Effective-dated revisions will preserve timetable history. The release will not provide arbitrary recurrence patterns, automatic holiday cancellation, room-conflict prevention, timetable optimization, or automatic attendance. Overlapping classes may be displayed without being automatically resolved. Student projection of recurring class timetables is outside this release; students will retain applicable coursework deadlines and authorized saved events.

Course-supervision pages will exclude student identities, individual grades, submissions, assessment attempts, answers, and teaching controls. The Super User will not receive grading, coursework-submission, assessment-taking, security-management, backup-management, or generic account-permission privileges. Dean and Program Head designations will remain distinct from the four primary system roles.

Audience-targeted calendar dispatch will mean visibility within the LMS, without implying external email, push delivery, or external-calendar synchronization. Converting a note to a calendar event, or an accessible event to a personal note, will create a snapshot. Later changes to either item will not continuously synchronize the copies. Generated class occurrences and coursework deadlines will remain distinct from saved calendar events.

Optional grade incentives will be subject to applicable academic rules and confirmed gradebook behavior. Their presence in the design will not imply institutional approval. Reporting, monitoring, and backup and recovery coverage will be limited to functions demonstrated in the evaluated version. Proposed additions will not be reported as completed or evaluated before implementation and testing establish their status.

The study will assess software quality within the selected setting and tasks. It will not determine long-term changes in grades, motivation, learning achievement, employability, or study habits. Findings from purposively selected respondents will be interpreted within the ICS context and will not represent all students or higher education institutions.

1.7 SIGNIFICANCE OF THE STUDY

The developed Colegio de Montalban Learning Management System (CdM LMS) is significant to the Institute of Computing Studies (ICS) as it provides an integrated environment for managing academic and technical learning activities across Web, Mobile, and Desktop platforms. The study may provide value to the following:

Institute of Computing Studies and Colegio de Montalban. The study may provide an institutional basis for organizing related academic information and examining whether a tailored LMS meets the needs of ICS. Its documented requirements and evaluation findings may support decisions about further development, maintenance, and possible adoption.

Students. The system is intended to provide a coordinated place for accessing coursework and feedback and managing personal study activities. Students may benefit from clearer access to their own academic information and opportunities for moderated peer support. These expected benefits remain distinct from measured improvements in academic achievement.

Instructors. The study may support more consistent handling of course resources, submissions, grading records, programming activities, and academic communication. Evaluation findings may identify interaction or workflow difficulties that require correction before broader use.

Academic supervisors and Program Heads. The proposed coordination functions may support authorized inspection of faculty and course information while preserving boundaries around student records and teaching controls. A defined supervisory role may help clarify which information is appropriate for academic coordination.

Administrators. The dedicated console is intended to support controlled account operations, system records, reporting, and supported continuity functions. Documenting access boundaries and audit behavior may assist in maintaining accountability for administrative actions.

Future researchers and developers. The study may provide a reference for examining an institutional LMS with complementary applications, computing-specific workflows, and limited academic supervision. Its limitations and evaluation findings may also identify questions requiring separate research, including longer-term educational outcomes and broader platform coverage.

1.8 GENERAL FEATURES OF THE SYSTEM / STUDY

The CdM LMS is built to handle the technical needs of the ICS department composed of interconnected Web, Mobile, and Desktop applications supported by a centralized database. The system provides role-based functionality for students, instructors, and administrators while maintaining synchronization of relevant academic and system information across its platforms. The following are the general features of the implemented system.

Core Functional Features

Academic and Course Management

• Courses, Sections, and Enrollment: Authorized users create and organize courses and sections. Students enter valid join codes to access their enrolled course workspaces.

• Course Workspace: Streams, classwork, people, and grades organize course information, activities, membership, and permitted academic records.

• Learning Materials and Annotations: Instructors upload and organize resources. Users view supported documents and add permitted highlights and sticky-note annotations.

• Assignments and Submissions: Instructors publish activities and deadlines. Students submit permitted work, subject to submission validation and applicable deadline restrictions.

• Course Announcements: Instructors publish course updates for the authorized audience, with relevant notifications available within the supported LMS workflow.

Grading and Academic Analytics

• Dynamic Gradebook: Instructors configure assessment components and weights, record scores, and obtain automatically computed grades using the configured grading rules.

• Student Grade View: Students view their own authorized grade information and available progress summaries.

• Visual Grade Analytics: Charts summarize existing grade and performance records for authorized students and instructors.

• CSV Grade Export: Instructors export permitted grade records for their assigned courses.

Programming and Collaborative Learning

• CodeLab: Students write, execute, and submit code in the browser. Instructors manage programming problems and predefined test cases used for assessment.

• Git-Integrated Submissions: Students submit applicable repository-based coursework for instructor review. Any repository authorization is handled separately from LMS login.

• Knowledge Exchange: Students post academic and technical questions, contribute answers, and share learning resources in a moderated environment.

• Faculty-Verified Answers and Reputation: Authorized instructors verify suitable answers and moderate content. Configured contribution activity earns reputation points separately from EXP.

Personal Study and Gamification

• Notes and Tasks: Users organize personal notes, task lists, and relevant academic reminders within the available client functions.

• Flashcards and Review: Students create and review flashcards, with configured review and mastery records supporting repeated study.

• Focus and Study Tracking: Focus timers and study-session records allow students to record and review their study activity.

• EXP, Levels, and Achievements: Eligible participation earns configured EXP and progression rewards, including levels, achievements, badges, and streaks.

• Department Leaderboards and Rewards: Students view permitted participation rankings. Instructors manage supported rewards; optional grade incentives remain subject to applicable academic rules.

Academic Coordination and Calendar

• Faculty Directory and Organizational Chart: Authorized users view faculty relationships, program affiliations, and Dean and Program Head designations. Faculty classification remains separate from course program classification.

• Academic Operations Hub: Super Users access authorized academic summaries, faculty filters, timetable information, permitted leadership assignments, and shared faculty announcements.

• Restricted Course Supervision: Authorized academic supervisors view permitted course metadata, instructor attribution, timetable, announcements, syllabus, and material summaries. Student identities, individual grades, submissions, attempts, answers, and teaching controls remain excluded.

• Weekly Schedules and Revision History: Authorized instructors and Administrators manage course schedules using selected weekdays, shared start and end times, date bounds, and Asia/Manila time. Effective-dated revisions preserve timetable history.

• Class Exceptions: Authorized users cancel, reschedule, or restore permitted future class occurrences without rewriting the entire recurring schedule.

• Calendar Views: Day, Week, Month, and Year views display authorized saved events, generated class occurrences, and coursework deadlines as distinct record types. Student recurring timetable projection is outside the stated release.

• Audience-Targeted Events: Saved events use Personal, Course, Institute Faculty, BSIT Instructors, or BSCpE Instructors audiences. Program Heads may publish to permitted faculty audiences; publication rights do not grant ownership of another creator’s content.

• Note–Event Snapshot Conversion: Users schedule an owned note as a private event or save an accessible event as a private note. The new item preserves accessible content at conversion time rather than continuously synchronizing later edits.

User Features

Role-Based Workspaces

• Student Workspace: Students access enrolled coursework, submit permitted activities, view personal grades, participate in Knowledge Exchange, and use available study tools.

• Instructor Workspace: Instructors manage assigned courses, materials, assessments, grades, submissions, and relevant academic communication. Mobile functions follow the coverage of the evaluated release.

• Super User Workspace: Super Users perform permitted academic oversight and coordination through the web, with access to their own private notes and events. Academic oversight does not grant teaching, grading, or generic administration privileges.

• Administrator Console: Administrators use the native Desktop Application for account approval, account status and roles, subject assignments, directories, department configuration, logs, reports, and supported continuity functions. Only Administrators provision or revoke Super User accounts.

• Program Head Designation: Eligible Instructor accounts receive defined program coordination responsibilities. Dean and Program Head remain academic designations rather than additional primary system roles.

System Capabilities

Cross-Platform Foundations

• Login and Registration: Users register and sign in through credential-based authentication. Verification and Administrator approval govern access to the assigned account role.

• Home Dashboard: The role-appropriate landing view presents available summaries, such as student progress and tasks, instructor course information, academic oversight summaries, or administrative status.

• Shared Data and Synchronization: The applicable Web, Mobile, and Desktop workflows use a shared information environment. Saved updates become available to other authorized interfaces through backend retrieval or synchronization, subject to connectivity and implemented update behavior.

• Automated Academic Processing: The system computes configured grades and progression values, generates applicable calendar occurrences, and filters information according to the user's permissions.

• In-LMS Updates: Relevant course and deadline notifications support academic tasks. Audience-targeted calendar dispatch controls LMS visibility without implying external email, push delivery, or external-calendar synchronization.

• Platform Help: An in-app help chatbot provides navigation and platform assistance.

Technology Integration

Connected Applications and Data Services

• Web Portal: The principal application supports academic workflows and the proposed coordination environment.

• Mobile Application: The complementary client supports selected academic and productivity functions according to role and release coverage.

• Desktop Application: The native Administrator Console supports permitted technical administration.

• Shared Backend and Database: Next.js backend services and a centralized PostgreSQL database connect the authorized applications and maintain applicable records.

• Programming Workflow Integration: Repository-based coursework and CodeLab connect to their respective academic activities. Supported providers, execution languages, and client technologies will follow the implementation documented in Chapter 3.

Security Features

Account and Information Protection

• Verified Account Access: Credential authentication, account verification, registration approval, and account-status controls govern LMS access.

• Server-Enforced Permissions: The backend checks session, role, institute, ownership, enrollment, and related-resource permissions before allowing actions or returning restricted information.

• Academic Privacy: Personal grades, notes, restricted records, and audience-targeted content remain accessible only through permitted workflows.

• Activity and Audit Records: Designated actions and changes are recorded to support administrative accountability, academic coordination review, and moderation oversight.

• Record Integrity: Database constraints, transactions, and version checks in the proposed expansion protect related records and concurrent updates.

• Repository Authorization: Access to applicable repositories follows its separately implemented authorization mechanism and does not replace general LMS authentication.

Unique or Innovative Features

ICS Workflow Integration

• Complementary Platform Responsibilities: Academic access, selected mobile support, and desktop administration operate around shared information with distinct role permissions.

• Computing Coursework Support: Separate CodeLab and Git-submission workflows support the different requirements of execution-based activities and repository-based coursework.

• Connected Learning Support: Course information, personal study tools, gamified progression, and moderated academic collaboration are organized within the ICS environment.

• Bounded Academic Oversight: A distinct Super User role supports coordination while restricted supervision preserves the boundary between academic leadership, teaching, and system administration.

• Schedule and Calendar Coordination: Revision history, occurrence exceptions, targeted events, and private snapshot conversion connect academic coordination and personal organization under defined access rules.

These features describe the proposed integration for ICS; their presence does not establish that comparable capabilities are absent from other configurable LMS products.

1.9 DEFINITION OF TERMS

The following terms are operationally defined according to their use in this study and are arranged alphabetically.

Academic Calendar - The interface presenting authorized saved events, generated class occurrences, and coursework deadlines through Day, Week, Month, and Year views.

Academic Operations Hub - The web interface through which a Super User views authorized academic summaries, faculty information, timetable filters, and course-supervision links.

Administrator - The system role responsible for authorized account, permission, configuration, log, reporting, and continuity operations through the native Administrator Console.

Agile Development - The iterative development approach in which requirements are implemented, tested, reviewed, and refined through successive project cycles.

Audit Log - A record of designated actions and changes used to support accountability, including authorized administrative and academic-leadership operations.

CdM LMS - The proposed interconnected learning management environment developed for the Institute of Computing Studies at Colegio de Montalban.

Centralized Database - The shared PostgreSQL data store supporting authorized academic and system records used by the applicable applications.

CodeLab - The in-browser programming environment for writing, executing, and submitting code assessed against predefined test cases.

Course Supervision - Restricted, read-only access to course metadata, instructor attribution, timetable, announcements, syllabus, and learning-material summaries.

Dynamic Gradebook - The grading module that uses configured components, weights, and recorded scores to compute and display authorized grade information.

Effective Date - The date from which a particular timetable revision governs the generation of applicable recurring class occurrences.

Experience Points or EXP - Configured participation points used for gamified progression and related rewards; they are not equivalent to course grades or forum reputation.

Faculty Directory - The authorized listing of ICS faculty, program affiliations, leadership designations, and relevant assigned-course summaries.

Faculty-Verified Answer - A Knowledge Exchange response marked by authorized faculty through the platform's answer-verification process.

Gamification - The use of configured points, levels, badges, achievements, streaks, and leaderboards to support participation in permitted LMS activities.

Git-Integrated Submission Engine - The workflow for repository-based coursework submission and authorized instructor review of applicable programming activities.

Institute of Computing Studies or ICS - The institute at Colegio de Montalban that defines the development context and research-evaluation scope of the study.

Instructor - The system role responsible for authorized teaching, assessment, grading, and course-related communication for assigned courses.

Knowledge Exchange - The moderated environment for academic and technical questions, answers, and resource sharing, with faculty verification and reputation mechanisms.

Learning Management System or LMS - The application environment used in this study to organize course resources, coursework, assessment, academic records, and related support workflows.

Program Head - An academic designation attached to an eligible ICS Instructor account for a specified program, with defined coordination privileges.

Reputation Points - Configured points associated with Knowledge Exchange contributions, maintained distinctly from EXP unless an explicit rule connects them.

Role-Based Access Control or RBAC - The mechanism that restricts system capabilities according to assigned roles, supplemented by institute, ownership, enrollment, and related-resource checks.

Schedule Exception - A record modifying an individual class occurrence through cancellation or rescheduling, or supporting restoration of its original occurrence.

Snapshot Conversion - Creation of a private note or calendar event from accessible source content at the time of conversion, without continuous synchronization.

Spaced-Repetition-Oriented Review - Flashcard review organized to support repeated study over time, without assuming an unverified scheduling algorithm.

Student - The system role permitted to access enrolled-course content, submit assigned work, view authorized personal academic records, and use available learning-support functions.

Super User - The system role for limited academic oversight and coordination, separate from Instructor and Administrator privileges.

Targeted Event - A saved academic event whose visibility and management are governed by its designated personal, course, institute-faculty, or program-instructor audience.

Timetable Revision - An effective-dated version of a course schedule used to preserve the historical representation of recurring class times.

Visual Grade Analytics - Descriptive charts and summaries derived from available grade and performance records, without predictive or AI-based inference.

CHAPTER 2: REVIEW OF RELATED LITERATURES

Digital technology has changed how educational institutions organize instruction and manage academic information. Learning Management Systems (LMS) provide a shared environment for distributing course materials, managing assignments, recording grades, and communicating with students. Access through different devices also creates opportunities to connect classroom activities with independent study. These functions provide the foundation for examining how an LMS can support academic work within higher education.

In the Philippine context, the studies reviewed in this chapter examine LMS adoption, user experience, accessibility, and institutional support. They also explore programming practice, gamification, personal learning strategies, and online collaboration. Their findings provide different perspectives on digital learning, including the difficulties users encounter and the conditions that support meaningful participation. Understanding these findings helps guide the selection and integration of functions appropriate to the Institute of Computing Studies (ICS) at Colegio de Montalban.

The project developers reviewed scholarly articles, conference papers, and system-based studies to establish the foundation for the proposed Learning Management System. Philippine and international studies are organized under seven topics: Multi-Platform Learning Management Systems; Course Management and Academic Communication; Programming Practice and

Automated Assessment; Git-Based Coursework Submission; Gamification and Student Engagement; Personal Study Tools and Self-Regulated Learning; and Online Collaboration and Knowledge Sharing. The discussion compares their findings, explains their connection to the proposed system, and identifies the research gap that the study will address.

2.1.1 Multi-Platform Learning Management Systems

Access through different devices should support useful academic tasks and an understandable learning environment. Alfalah (2023) surveyed 258 students from Hail University, Aljouf University, and Northern Border University in Saudi Arabia about mobile LMS adoption, including the Blackboard application. Performance expectancy, effort expectancy, and learning intention were associated with behavioral intention to use the system. In the Philippines, Dela Rosa (2023) examined Canvas Free for Teachers with 132 third-year BSIT students taking Web Systems and Technologies 2 at Bulacan State University. Students evaluated its lesson modules and assessments favorably. Both studies connect adoption with users' experiences of academic functions, although Alfalah examined mobile-use intentions and Dela Rosa examined acceptance within a specific programming course.

Philippine development studies provide more direct examples of designing learning access across devices. Quimod et al. (2023) developed e-Modyul, a mobile and web LMS for Technical-Vocational-Livelihood education, and evaluated it with 10 teachers and 20 students. Its approach supports accessing lessons and assessments through different operating systems. Ramos et al. (2024), researchers associated with the University of the Philippines Open University (UPOU), developed a Filipinized LMS responsive to Filipino nursing educators and learners. Their design considered connectivity limitations, multimodal learning materials, communication, and collaboration. These studies share an emphasis on access, but e-Modyul focuses on device coverage, while the Filipinized LMS emphasizes adaptation to its users and educational context.

Platform access also depends on the quality of everyday workflows. Llamas et al. (2025) examined the Moodle-based LMS at Mindanao State University–Iligan Institute of Technology (MSU–IIT), involving six academics and six fourth-year students. They identified concerns in usability, communication, course building, monitoring and assessment, content management, and technical support. Their findings qualify the favorable acceptance evidence from Dela Rosa: providing an LMS does not remove all difficulties in using it.

Taken together, these five studies support designing complementary interfaces around the tasks and access conditions of the intended users. For CdM LMS, the web application will provide the principal academic environment, the mobile application will support selected student and instructor functions, and the desktop application will serve the Administrator. Multi-platform access therefore concerns coordinated responsibilities and consistent shared records rather than identical functionality on every device. The reviewed studies inform this arrangement, while its suitability for ICS will require evaluation of the implemented applications.

2.1.2 Course Management and Academic Communication

Course management connects learning materials, activities, academic records, and communication within a clear workflow. Peralta (2024) developed a course management system for Centro Escolar University and evaluated it with 100 students and eight faculty members. It organized materials, announcements, assignments, quizzes, and grades and included Short Messaging Service notifications. Llamas et al. (2025), however, identified course-building, content-management, communication, and assessment concerns in the MSU–IIT LMS. These studies suggest that organizing functions within one platform is useful, but users must still be able to complete their tasks without unnecessary difficulty.

The communication experience also varies across educational settings. Acala and Dimasacat (2023) examined perceptions of 74 private senior high school students in Manila and reported favorable views of LMS-supported interaction and learning. They recommended training to help

students and teachers use the available functions. Laspiñas and Almerez (2024), in contrast, surveyed 33 business graduate students at an unnamed university in Cebu City and reported generally neutral evaluations of the LMS's pragmatic and hedonic qualities. Their findings address different populations, yet both indicate that communication and task support should be assessed through users' experiences rather than inferred from the presence of features.

Institutional support affects whether these workflows can be maintained. Guico (2026) examined 182 faculty members and 29 administrators from 11 higher education institutions in Batangas, including Batangas State University, Lipa City College, and Rizal College of Taal. Reported concerns included preparation time, resistance to change, access to information and communication technology resources, and continuing training. Francisco (2026) examined remote learning management at the University of Rizal System and likewise identified connectivity, digital-resource, technological-competence, engagement, and assessment-integrity challenges. Both studies place course management within a wider environment of faculty support and institutional coordination.

Across these six studies, effective academic communication depends on understandable course organization, appropriate user preparation, and the support needed to sustain the system. This evidence informs CdM's course materials, announcements, submissions, feedback, dynamic grading, and descriptive grade analytics. It also supports considering faculty organization, schedules, and targeted academic events as part of course coordination. Peralta's SMS function remains a comparison feature of that system; the proposed CdM workflow uses its own documented notification functions. Requirements gathering at ICS will establish the local difficulties to be addressed.

2.1.3 Programming Practice and Automated Assessment

Programming practice requires suitable exercises and feedback that explains performance against clear criteria. Messer et al. (2024) reviewed 121 papers on automated grading and feedback tools for programming education. Many tools evaluated correctness through dynamic methods such

as unit tests, while readability, maintainability, and documentation received less attention. Novak and Kermek (2024), from the University of Zagreb in Croatia, examined automation for complex programming assignments using unit testing and custom scripts. Both studies support automating selected checks while retaining instructor judgment for aspects that those checks cannot adequately assess.

Philippine research adds considerations about introductory practice and access. Asuncion et al. (2022), affiliated with Ateneo de Manila University and the National Olympiad in Informatics–Philippines, presented the ABC Workbook and AutoJudge++. The project combines introductory Python exercises with an offline desktop checker that compares outputs against test cases. Classroom effectiveness evaluation remained future work. Arada (2025) examined Tinkercad with 42 first-year Computer Technology students at a Philippine state university. The platform received a below-average System Usability Scale score of 58, despite many students expressing willingness to use it. These studies concern different tools, but together they show that opportunities to practice should be considered alongside exercise design, usability, and access conditions.

Assessment can also concern development processes beyond the output of an individual program. Fernandez-Gauna et al. (2023), from the University of the Basque Country in Spain, described automated feedback and assessment for team-coding assignments in a DevOps context. Their approach used team and individual indicators and periodic reports to support monitoring. Compared with the test-case approaches examined by Messer et al. and Asuncion et al., this work illustrates a broader form of assessment involving collaborative development activity. It does not establish that development indicators can replace assessment of the submitted program.

The five studies support matching automated feedback to the assignment and preserving the limits of what is measured. CdM's CodeLab will evaluate programming work against predefined test cases and provide understandable results. Its checks will concern documented expected behavior rather than complete program quality. Git-based coursework review will remain a separate workflow. The related studies therefore guide test design, feedback presentation, and usability without

expanding CodeLab into automated code repair, artificial intelligence grading, or DevOps-based team assessment.

2.1.4 Git-Based Coursework Submission

Repository-based coursework can connect code submissions with an organized record of development and instructor review. Schauer et al. (2024), from Heriot-Watt University in the United Kingdom, developed a framework integrating Canvas LMS and GitLab. They identified fragmented information and poor user experiences when the platforms were used independently and implemented eight applications to connect their workflows. Orvalho et al. (2024), affiliated with the University of Lisbon and Czech Technical University in Prague, presented GitSEED, a GitLab-backed automated assessment tool. Both approaches connect academic work with version control, although Schauer et al. emphasize LMS integration and GitSEED provides assessment and feedback through repository workflows.

Evidence about students' use of repositories complements these system-development approaches. Patani et al. (2024) examined GitHub use in a software engineering course with 319 undergraduate students, using surveys and repository records such as commits, pull requests, and issues. They reported improvements in collaborative development and teamwork after the course experience. Fernandez-Gauna et al. (2023) used repository-related development indicators to provide feedback on team-coding assignments. Together, these studies show how repositories can make development activity available for educational review, but their measures extend beyond recording a final coursework submission.

Philippine evidence places GitHub within a wider teaching approach. Bernardez et al. (2026), from the University of the Philippines Cebu, examined Design Thinking and Agile-Scrum across two software engineering courses involving 65 students. Their findings describe GitHub and Trello as supporting version control and task coordination, alongside challenges involving time pressure and uneven workloads. Since GitHub formed part of a combined instructional approach, the study cannot

isolate its contribution to learning outcomes. It nevertheless demonstrates the use of repository tools in Philippine computing coursework.

These five studies support examining how academic instructions, repository records, and instructor feedback can be connected. For CdM LMS, Git authorization will support the documented repository submission and review process, while LMS account access will follow its separate authentication and approval rules. Repository submissions will not automatically become CodeLab assessments or commit-based academic grades. The project's contribution is to evaluate its particular coursework workflow for ICS, informed by the integration concerns identified by Schauer et al. and the educational repository use described in the other studies.

2.1.5 Gamification and Student Engagement

Gamification can encourage participation when rewards are connected to meaningful learning activities. Ahmed et al. (2025) evaluated a gamified, mobile, cloud-based LMS with 60 second-year students taking Object-Oriented Programming at Mansoura University in Egypt. The experimental group obtained higher post-test achievement and engagement scores than the comparison group. Jaramillo-Mediavilla et al. (2024) synthesized gamification research and reported favorable findings in parts of the literature while identifying gaps in examining motivation and academic performance together. These studies support investigating gamification, but the outcomes of a particular intervention should not be treated as a guaranteed effect of every reward system.

The design of participation incentives matters. Jack et al. (2025) examined points and leaderboards in a flipped statistics classroom and discussed benefits for motivation and engagement alongside the risks of superficial engagement and demotivation. Philippine evidence from Fojas (2022) examined gamification in Moodle courses at UPOU, comparing a gamified cohort with earlier non-gamified cohorts and collecting student feedback. The reported differences favored the gamified course, but forum-grading arrangements also differed between cohorts, and some students described

anxiety or limited interest in rewards. Both studies show why participation counts and positive feedback need to be interpreted in relation to the design of the activities.

Benben and Bug-os (2022) studied Quizizz-based formative assessment with Grade 10 physics students at Central Mindanao University Laboratory High School in Bukidnon. Achievement and motivation increased after the intervention, although the reported learning gain was low. Its one-group pretest-posttest design and secondary-school setting differ from Ahmed et al.'s computing-course comparison. The studies therefore provide complementary evidence about possible motivational benefits while leaving questions about transfer to other subjects and populations.

Across these five studies, gamification is most relevant when its rules encourage participation without obscuring academic expectations. CdM LMS will use EXP, levels, achievements, streaks, and leaderboards under defined rules. These records will remain separate from academic grades and Knowledge Exchange reputation. The literature supports the rationale for these mechanisms and the need to examine users' experiences, while the proposed descriptive software-quality evaluation will not establish their long-term causal effects on achievement or motivation.

2.1.6 Personal Study Tools and Self-Regulated Learning

Personal study tools can support planning, active review, and reflection when learners use them purposefully. Du et al. (2025) examined an LMS enhanced with self-regulated learning support among 69 management undergraduates taking mathematics at an unnamed large public university in China. Students reported stronger self-regulation, but an examination advantage appeared on the first test and not the second. Heikkinen et al. (2023), in a review of 56 studies, found that only 46 percent of the reviewed learning-analytics interventions reported positive effects on learning. Both studies indicate that learning support deserves investigation, while access to monitoring tools alone does not ensure improved outcomes.

Students' interactions with LMS tools help explain differences in how learning support is used. Çakiroğlu et al. (2024) examined a third-year Operating Systems and Applications course at an

unnamed public university in Turkey. They identified different learner profiles and relationships between assessment interactions and self-regulation skills, including goal setting, self-evaluation, and help-seeking. In the Philippines, Funa et al. (2023) surveyed 301 preservice teachers in the Bicol region and supplemented the survey with interviews. Participants reported relatively strong environment structuring and goal setting but also difficulties with task strategies, time allocation, priorities, and interaction. These studies connect self-regulation with both individual learning practices and the conditions in which academic tasks are completed.

Active review provides another form of personal learning support. Pan et al. (2023) compared user-generated and premade digital flashcards in six experiments involving delayed tests. Creating flashcard content generally supported memory and, in several conditions, application performance. Unlike the broader LMS and analytics studies, this research examines a specific review activity. It offers a direct basis for considering student-created flashcards alongside the organization and reflection functions discussed by Du et al., Çakiroğlu et al., and Funa et al.

Together, the five studies support making planning, review, and progress information available within the learning environment. For CdM LMS, personal notes, tasks, flashcards, review and mastery records, and focus-study records will provide ways to organize and revisit academic work. Descriptive displays can help users inspect their saved records, but their presence will not be treated as evidence of improved self-regulation. The implemented functions will require evaluation for usability and correct behavior, while their educational effects remain a possible direction for later research.

2.1.7 Online Collaboration and Knowledge Sharing

Online knowledge sharing depends on the quality of interaction and the way discussions are organized. Oyarzun and Martin (2023) reviewed 63 studies on online learner collaboration and examined technologies, instructional design, facilitation, and outcomes. Their review connects collaboration with the conditions supporting participation. Bach and Thiel (2024) surveyed 298

undergraduate teacher-education students in asynchronous university courses and examined cognitive, metacognitive, and relational dimensions of interaction. Their findings concern associations with self-reported learning gains and satisfaction rather than causal effects. Both studies indicate that a shared online space should be assessed through the exchanges it supports.

Structured discussions and instructor involvement provide more specific guidance. Mudau and Van den Berg (2023), from the University of South Africa, examined online discussions involving 11 master's students in education. They identified the importance of clear guidelines, structured questions, and opportunities for discussion within a community of inquiry. Almodiel (2022) analyzed knowledge construction in asynchronous discussions within a graduate colloquium at UPOU. The study used the Interaction Analysis Model to examine the exchanges through which learners shared and developed knowledge. These studies support considering discussion structure and facilitation alongside opportunities for students to contribute.

Participation measures should also reflect interaction between users. Kumi (2023) examined discussion-forum exchanges in an undergraduate class with 31 students using social network analysis. The analysis revealed structural and relational aspects of participation that simple counts or self-reports could overlook. This complements Almodiel's examination of knowledge construction: both approaches look beyond the number of messages to consider how students interact. Their analytical methods are research approaches rather than functions automatically required in the proposed CdM system.

The five studies support designing Knowledge Exchange as an organized space for academic questions, answers, and resource sharing. Moderation and Faculty-Verified Answers will provide project-specific ways to manage contributions and identify instructor-reviewed responses. Reputation will describe forum participation under its configured rules and remain separate from EXP and academic grades. The literature provides reasons to examine participation, facilitation, and contribution quality, while the suitability of CdM's verification and reputation mechanisms will

require its own evaluation. The proposed forum supports asynchronous exchanges; the reviewed studies do not require adding live meetings or collaborative document editing.

2.2 Supporting Quality Framework and Local Study Context

ISO/IEC 25010:2023 provides the product-quality model supporting the study's selected evaluation characteristics: functional suitability, interaction capability, security, performance efficiency, reliability, and compatibility (International Organization for Standardization, 2023). The standard is a supporting framework rather than one of the five studies counted within each topic. Questionnaire feedback will describe respondents' evaluations, while calculation checks, access tests, and integration verification will provide evidence about implemented behavior.

The reviewed Philippine studies provide context from computing courses, institutional LMS use, programming practice, gamified learning, and online discussions. Francisco (2026) adds evidence from the University of Rizal System, bringing access and coordination concerns closer to the geographical setting of Colegio de Montalban. Findings from these institutions do not establish the conditions within ICS. Requirements gathering and system evaluation will identify the needs and experiences of the intended BSIT and BSCpE users.

2.3 Related Systems

The following systems provide documented examples of course organization, progress information, and repository-based workflows. The comparison uses the capabilities described in the cited sources rather than an exhaustive feature audit. It does not assume that these platforms lack other functions or cannot be extended.

2.3.1 Moodle

Moodle's publication on LMS 4.5 described improvements to assignment interaction and course subsections for organizing content (Moodle, 2024). Llamas et al. (2025) examined its institutional use at MSU–IIT, while Fojas (2022) and Almodiel (2022) examined gamification and

discussions in the UPOU context. These sources show that Moodle can support several forms of academic work. For CdM LMS, the relevant comparison concerns the organization of coursework and the experience of completing institutional tasks.

2.3.2 Google Classroom

Mennuti (2024) described targeted assignments, rubrics, and analytics for grades, completion rates, and missing work in Google Classroom. The analytics described were associated with specified education editions or upgrades. These functions provide comparison points for assignment management and descriptive progress information. Their documentation does not establish the quality of the proposed CdM implementation or demonstrate that it will outperform Classroom.

2.3.3 Canvas and GitLab Integration

Schauer et al. (2024) implemented a framework connecting Canvas and GitLab to reduce fragmented workflows and support feedback. Its eight applications demonstrate that existing LMS and version-control platforms can be connected for educational use. This is a direct comparison for the integration concern in CdM's Git submission workflow. The technologies and automation described in that framework remain distinct from CdM's specified implementation.

2.3.4 GitSEED

Orvalho et al. (2024) presented GitSEED as a GitLab-backed automated assessment tool supporting programming and software engineering education. It uses repositories as part of submission and feedback workflows. Its approach provides a comparison for connecting coursework with version control, but its automated assessment capabilities extend beyond CdM's separate repository-submission and instructor-review process.

2.5 Synthesis of the Literature

The literature connects LMS adoption with useful academic functions, understandable interaction, and conditions supporting continued use. Alfalah (2023) and Dela Rosa (2023) address acceptance, while Quimod et al. (2023), Ramos et al. (2024), and Llamas et al. (2025) provide development and user-experience perspectives. Peralta (2024), Guico (2026), and Francisco (2026) further connect coursework with communication and institutional support. These studies support examining connected workflows and appropriate access across CdM's complementary applications.

Programming and repository research address related but distinct academic processes. Messer et al. (2024), Asuncion et al. (2022), and Novak and Kermek (2024) inform checks against specified programming criteria. Schauer et al. (2024), Orvalho et al. (2024), and Patani et al. (2024) inform repository use and integration. The comparison supports keeping CodeLab's test-case assessment separate from Git-based coursework review, with instructors retaining responsibility for judgments outside the automated checks.

Gamification, personal study support, and collaboration concern different aspects of participation. Ahmed et al. (2025), Jack et al. (2025), and Fojas (2022) provide evidence about rewards and engagement. Du et al. (2025), Heikkinen et al. (2023), Funa et al. (2023), and Pan et al. (2023) address regulation and active review. Oyarzun and Martin (2023), Almodiel (2022), and Mudau and Van den Berg (2023) emphasize interaction and facilitation. Their combined implication is to provide learning support with clear purposes and evaluate how the intended users experience it.

For the present study, these findings inform the development and descriptive evaluation of an LMS configured for ICS. Evidence from other institutions provides design guidance and comparison points. The study will establish its own findings about software quality and workflow suitability, while recognizing that user ratings and functional verification do not establish long-term educational effects.

2.6 Research Gap

The reviewed studies address digital learning and institutional LMS utilization but do not evaluate the proposed CdM LMS configuration for ICS at Colegio de Montalban. The research gap is contextual and evaluative: evidence is needed about how its combined workflows, roles, and complementary applications operate in the intended setting. One part of the gap concerns functional integration. CdM LMS will combine routine academic management with separate CodeLab and Git-submission workflows, learning-support tools, and moderated collaboration. Its evaluation must determine whether the selected functions produce the expected results and interact consistently within a shared information environment. This is a system-specific question; it is not a claim that no existing platform can provide similar integration. A second part concerns academic coordination and information boundaries. The proposed expansion introduces an academic Super User, restricted course supervision, faculty organization, effective-dated timetable revisions, class exceptions, and audience-targeted events. The reviewed literature provides organizational reasons to consider coordination, but it does not establish whether CdM's particular implementation preserves schedule behavior, correct audiences, and restricted information access. These properties require requirements-based testing and evaluation of the implemented release. A third part concerns role-appropriate access across applications. The web environment, complementary mobile client, and native Administrator Console have distinct responsibilities. The study will examine consistency in their applicable shared records and the quality of the tasks each interface supports, without assuming universal feature parity. Findings will identify the suitability and limitations of this configuration for ICS. The study will address these gaps through development, documented verification, and descriptive software-quality evaluation, producing evidence about the suitability and limitations of the ICS implementation.

CHAPTER 3: METHODOLOGY

3.1 RESEARCH DESIGN

The study will employ a quantitative descriptive design through survey questionnaires. Descriptive research aims to describe the characteristics of a population, situation, or phenomenon systematically and accurately (McCombes, 2023). Through this design, the project developers will obtain quantitative data from respondents' feedback using two sets of questionnaires: one for eligible BSIT and BSCpE students and instructors of the Institute of Computing Studies at Colegio de Montalban, and another for qualified IT experts. Both instruments will be aligned with the six selected characteristics of the ISO/IEC 25010:2023 Software Product Quality Model: functional suitability, interaction capability, security, performance efficiency, reliability, and compatibility.

The questionnaires will undergo review by the capstone adviser and qualified educator-validators to assess their clarity, relevance, and appropriateness before administration. Respondents will evaluate the functions they have used through tasks appropriate to their roles, while IT experts will conduct the applicable technical checks. This approach will enable the project developers to determine and describe the software quality of the Learning Management System in supporting academic management, learning activities, and coordination within ICS.

3.2 System and Project Overview

The proposed project, "Learning Management System for the Institute of Computing Studies at Colegio de Montalban," will provide a centralized environment for managing academic activities and supporting student learning. The system will serve students and instructors of the Bachelor of Science in Information Technology (BSIT) and Bachelor of Science in Computer Engineering (BSCpE) programs. It will consist of a web application, a mobile application, and a desktop application connected through a shared backend and database.

The web application will serve as the primary platform for course management, academic communication, assessment, and student progress monitoring. Instructors will be able to organize

course materials, create assignments, publish announcements, review submissions, configure grading components, and view descriptive grade analytics. Students will be able to access enrolled courses, retrieve learning materials, submit coursework, and monitor their academic records. The system will also include CodeLab for programming activities evaluated against predefined test cases and a separate Git-based coursework submission workflow for repository-linked submissions and instructor review.

The mobile application will provide access to selected academic functions and personal study tools. These tools will include notes, task management, flashcards, focus timers, and study-session records. Gamification features, such as experience points (EXP), levels, badges, achievements, streaks, and leaderboards, will support participation and engagement. These indicators will remain distinguishable from students' academic grades.

The system will also provide a Knowledge Exchange forum where students and instructors can ask questions, discuss academic concerns, and share learning resources. Moderation and faculty-verified answers will support the organization and credibility of shared content. Academic coordination features will include schedules, calendar views, targeted events, and authorized course oversight. The desktop application will function as an administrator console for account management, system configuration, logs, reports, and supported maintenance activities.

Following development and testing, the researchers will evaluate the implemented system through structured questionnaires and relevant technical checks. The evaluation will address the selected ISO/IEC 25010 software quality characteristics: functional suitability, interaction capability, security, performance efficiency, reliability, and compatibility. Respondents will assess only the functions available in the evaluated release and relevant to their assigned roles.

3.3 Respondents

The respondents will consist of selected ICS students, ICS instructors, and qualified IT

experts. These groups will provide complementary assessments of the system's academic usefulness, user experience, and technical quality. The student respondents will be currently enrolled BSIT and BSCpE students at Colegio de Montalban who have previously used at least one Learning Management System. Before answering the questionnaire, they will receive an orientation and perform assigned activities using the proposed system. These activities may include accessing course materials, submitting coursework, viewing personal grades, using study tools, and participating in the Knowledge Exchange forum. Their responses will describe their experience with the student functions they have directly used.

The instructor respondents will be faculty members currently teaching BSIT or BSCpE courses in the ICS who have previous experience using an LMS. They will perform relevant activities, such as managing course materials, creating assessments, reviewing submissions, encoding grades, communicating with students, and monitoring academic progress. Instructors with an assigned academic designation may also evaluate the corresponding coordination or oversight functions when these are included in the evaluated release.

The IT expert respondents will be selected practitioners or faculty members with relevant experience in software development, software testing, database management, cybersecurity, or system administration. They will evaluate technical functions within their areas of competence using the questionnaire and designated test activities. Where necessary, controlled test accounts will provide access to administrative functions without requiring unrestricted access to actual student records.

The three faculty validators and the research adviser who will review the questionnaires will perform a separate instrument-validation role. Their review will not automatically make them evaluation respondents unless they independently meet the respondent criteria.

The final number of respondents will be established after confirming the eligible population

and evaluation arrangements. The researchers will report the actual number of participating students, instructors, and IT experts separately.

3.4 SAMPLING TECHNIQUE

The study will use criterion-based purposive sampling, a non-probability approach in which participants are deliberately selected according to predefined requirements relevant to the research. Memon et al. (2025) explain that criterion sampling requires clear and justified selection criteria aligned with the study's purpose and may be applied in quantitative research. This technique is appropriate because the study requires assessments from individuals who can directly evaluate the proposed LMS within its intended setting. Students will assess learning and submission activities, instructors will assess course management and grading workflows, and IT experts will assess technical quality. Enrollment in the institution alone will not be sufficient to provide an informed assessment of every system function.

The researchers will establish separate criteria for the three respondent groups. Student respondents must be enrolled in the BSIT or BSCpE program of the Institute of Computing Studies at Colegio de Montalban and have prior experience with at least one LMS. Instructor respondents must teach within ICS and have prior LMS experience relevant to their academic responsibilities. IT expert evaluators must possess relevant experience in software development, software testing, database management, security, or systems administration. All participants must voluntarily agree to participate, and they will perform the assigned evaluation tasks before providing their assessments.

These criteria will connect participant selection to the study's software-quality questions. Students will provide feedback on permitted academic and learning-support activities, instructors will assess teaching and course-management workflows, and IT experts will examine applicable technical behavior. Selection will therefore be based on participants' suitability for the assigned evaluation rather than their availability alone. The findings will describe the selected participants' assessments

and the evaluated system within the ICS setting, without claiming statistical representation of the entire population.

3.5 INSTRUMENTS AND METHODS FOR DATA COLLECTION

The researchers will use survey questionnaires as the primary instruments for collecting quantitative feedback about the Learning Management System. Survey research supports the collection of numerical responses that can be summarized through frequencies, averages, and patterns (McCombes, 2023). The questionnaires will use consistent instructions and structured statements to obtain comparable feedback about the functions available to each respondent group.

The researchers will develop two sets of questionnaires aligned with the six selected characteristics of the ISO/IEC 25010:2023 Software Product Quality Model: functional suitability, interaction capability, security, performance efficiency, reliability, and compatibility. The first set will be administered to qualified IT experts to assess applicable technical functions and workflows. Their questionnaire responses will be supported by technical checks involving grade computations, access permissions, record consistency, response times, failure handling, and information exchange among the connected applications. This combination will provide technical evidence alongside expert assessments.

The second set will be administered to eligible ICS students and instructors after they perform tasks appropriate to their roles. Students will evaluate available coursework, submission, personal grade, study, and collaboration functions. Instructors will evaluate assigned course-management, assessment, grading, and communication functions. Respondents will assess only the functions they have used through the Web Portal and supported Mobile Application. The administrator-only Desktop Application will be examined by qualified evaluators using authorized Administrator test accounts. Role-specific questionnaire sections will prevent participants from rating functions outside their access or experience.

Both questionnaires will use a four-point Likert-type response scale: Strongly Agree (4), Agree (3), Disagree (2), and Strongly Disagree (1). Statements will describe specific, observable system behavior and will address one aspect at a time. A separate “Not applicable or not observed” option will be provided for functions that a respondent cannot assess. This option will not receive a numerical score and will be excluded from the calculation of mean ratings.

Before administration, the questionnaires will undergo review by the research adviser and three qualified faculty validators from Colegio de Montalban. They will examine the relevance, clarity, and alignment of the items with the Statement of the Problem, selected quality characteristics, and respondent responsibilities. The researchers will revise the instruments based on their recommendations and conduct a pilot with eligible participants outside the main evaluation sample to identify unclear statements and difficulties with the task instructions. Reviewer approval will support content validity, while any assessment of internal consistency will require an appropriate analysis of actual pilot responses.

During data gathering, participants will receive an orientation, provide informed consent, and complete the assigned evaluation tasks before answering the questionnaires. The researchers will record the evaluated release, applicable platform, task completion, and encountered difficulties. Questionnaire responses and technical results will be organized and analyzed separately, then considered together when answering the software-quality questions in the Statement of the Problem.

3.6 HARDWARE REQUIREMENTS

The development and evaluation of CdM LMS will require development workstations, web-access computers, mobile test devices, an Administrator workstation, hosting resources, and network connectivity. The following specifications provide estimated baseline and recommended configurations. The researchers will verify these configurations using the actual evaluation release before reporting them as supported requirements.

Table ?. Estimated hardware requirements for development and client access

Equipment	Estimated baseline configuration	Recommended configuration	Purpose
Development workstation	Modern 64-bit, four-core processor; 16 GB RAM; SSD with at least 50 GB available space	Six-core or better processor; 32 GB RAM; 100 GB available SSD space	Develop and build applications, run browser tests, and operate local services
Web client computer	Modern dual-core processor; 4 GB RAM; approximately 1 GB available space for browser cache and temporary downloads; 1366 × 768 display	Four-core processor; 8 GB RAM; 5 GB available space; 1920 × 1080 display	Access coursework, grades, study tools, and authorized coordination functions
Android evaluation device	Device compatible with the evaluation build; approximately 4 GB RAM and 1 GB available storage	6 GB RAM or more and 2 GB available storage	Evaluate enabled academic and productivity functions
iOS evaluation device, if included	Compatible iPhone or iPad with approximately 1 GB available storage	A currently supported device with 2 GB available storage	Evaluate the iOS build, only if it is included in the tested release
Administrator workstation	Modern 64-bit dual-core processor; 4 GB RAM; 2 GB available storage; 1366 × 768 display	Four-core processor; 8 GB RAM; 5 GB available storage; 1920 × 1080 display	Run the native Administrator Console and retain permitted exports

Network equipment	Reliable router/access point and stable internet connectivity	Wired connections for laboratory computers and sufficient wireless capacity for the evaluation group	Connect clients to backend, database, storage, and external services

Development storage estimates include dependencies, build outputs, and test artifacts. Additional space will be required for Android emulators, container images, and larger datasets. A 32 GB development workstation is preferable when running an emulator, Docker services, the application, and browser tests simultaneously. The available client storage figures are planning allowances for installation, caching, downloads, and exports; they are not measured application sizes.

3.7 SOFTWARE REQUIREMENTS

CdM LMS uses a Next.js web application and backend, a centralized PostgreSQL database accessed through Prisma, an Expo/React Native mobile application, and an Electron Administrator Console. The researchers will retain a release configuration record identifying the exact dependency versions, hosting configuration, enabled modules, and supported client builds.

Table ?. Software requirements and current implementation technologies

Software or service	**Current choice or proposed configuration**	**Purpose**
Development operating system	Windows 11, or a compatible supported macOS/Linux development environment	Source editing, builds, and testing; platform-specific build requirements still apply

Node.js	Supported Node.js 22 LTS patch, **22.12 or later**	Run development tools and the Next.js backend; record the exact tested patch
pnpm and Turborepo	pnpm **10.32.1**; repository-managed Turborepo dependency	Manage monorepo dependencies and application tasks
Web framework	Next.js **16.2.0**	Web interface, server rendering, API handlers, and server actions
Web interface libraries	React **19.2.4**, TypeScript **5.9.3**, and Tailwind CSS **4.2.2**	Interface implementation, typing, and styling
Database	PostgreSQL; **version 16 is a proposed evaluation baseline**, while the deployed LMS version must be verified	Maintain relational academic and system records
Database access	Prisma **7.9.1**, PostgreSQL driver adapter	Database access, schema generation, and reviewed migrations
Mobile framework	Expo **55.0.31**, React Native **0.83.10**, and Expo Router	Selected mobile academic and productivity workflows
Desktop framework	Electron **41.0.3** and Electron Builder	Package and run the native Administrator Console

Web browsers	Current stable Chrome and Edge as initial test targets; Firefox and Safari were included	Access the Web Portal; document exact tested versions
Programming execution	Configured Judge0-compatible service; deployed service version and enabled languages to be verified	Execute programming submissions and predefined test cases
Version control and coursework repositories	Git and the existing GitHub integration	Development version control and authorized repository coursework
Persistent file service	Configured object storage; the repository includes Vercel Blob integration	Retain uploaded resources and applicable snapshots
Email service	Configured SMTP/email service where verification or recovery is enabled	Deliver supported account-related messages
Testing tools	Playwright **1.63.0**, TypeScript checks, Node.js tests, and repository harness tools	Functional, authorization, integration, and browser verification
Performance observation	Browser developer tools, application logs, database observations, and a selected load-testing tool	Measure timing, errors, resource use, and workload behavior

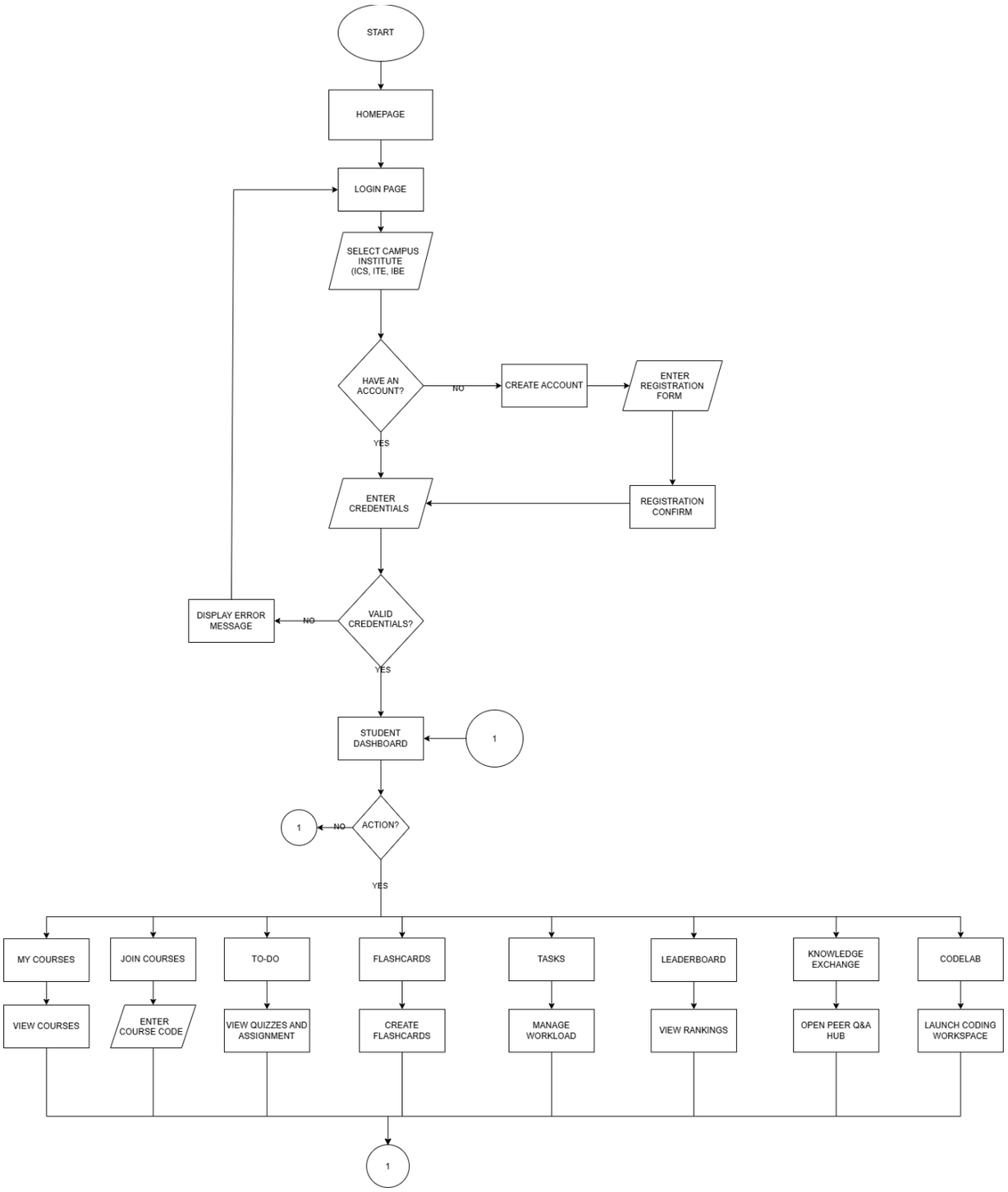
Analysis software	Proposed: Microsoft Excel or LibreOffice Calc; additional statistical software if required	Tabulate responses and calculate approved descriptive statistics

Node.js 22 is consistent with the repository's current CI configuration. Its exact patch should satisfy all dependencies, including Prisma's installed requirement. Node.js identifies versions 22 and 24 as LTS releases; an unsupported runtime should not be selected merely because an older framework minimum permits it.

Development tools such as Node.js, pnpm, Prisma, Docker, and Playwright are **not installation requirements for ordinary students or instructors**. Web users need a compatible browser; mobile and desktop users need the appropriate client build.

3.8 Flowchart

Student's View:



STUDENT VIEW

Students can access the Learning Management System (LMS) through the home page to the login page. Prior to logging in, students are required to select their institute from the campus and go through the process of account authentication.

1. User Authentication

On arriving at the login page, the LMS checks if there is any account for the student.

A. In Case of Non-Existence of Account for the Student

The student needs to register the account by filling up the required fields of the form. The registration confirmation will be done through the processing of the form. Once the registration process is completed, students can log in through the provision of their login credentials.

B. In Case of Existence of Account for the Student

The students can skip the registration process and provide their login credentials directly.

2. Login Verification

Upon providing the credentials, the system will verify the provided information.

the credentials are not valid:

- An error message will be shown by the system.
- Redirecting students back to the login page.

the credentials are valid:

- The system will provide access to students' accounts.
- Redirecting the students to the Student Dashboard.

3. Student Dashboard

After a successful login process, students are able to get into the Student Dashboard and choose from the features of LMS based on their academic and learning purposes.

A. My Courses

Students may choose My Courses to see the list of the courses that they have registered for. This allows them to access course-related information.

B. Join Courses

Students may choose Join Courses and register for a new course by providing a course code. The system accepts the provided code and registers students to the chosen course.

C. To-Do

Students can use the To-Do feature to see their quizzes and assignments. As a result, they will be able to keep track of their school responsibilities and remember what they have to do.

D. Flashcards

Students can use the Flashcards feature to make flashcards for reviewing their lessons and for organising their study material, and using these flashcards is helpful when studying on their own.

E. Tasks

Students can use the Tasks feature to carry out their work and look after their school duties. This helps them to keep up with the activities and responsibilities which they have planned.

F. Leaderboard

Students can choose to select the Leaderboard feature in order to see the rankings. This feature shows where they stand in the system's ranking list.

G. Knowledge Exchange

Students can use Knowledge Exchange to ask questions and start conversations. The feature gives them a place in which they can share information, talk about their ideas and discuss matters related to school.

H. CodeLab

Students have the possibility of accessing CodeLab in order to start up the coding workspace. The

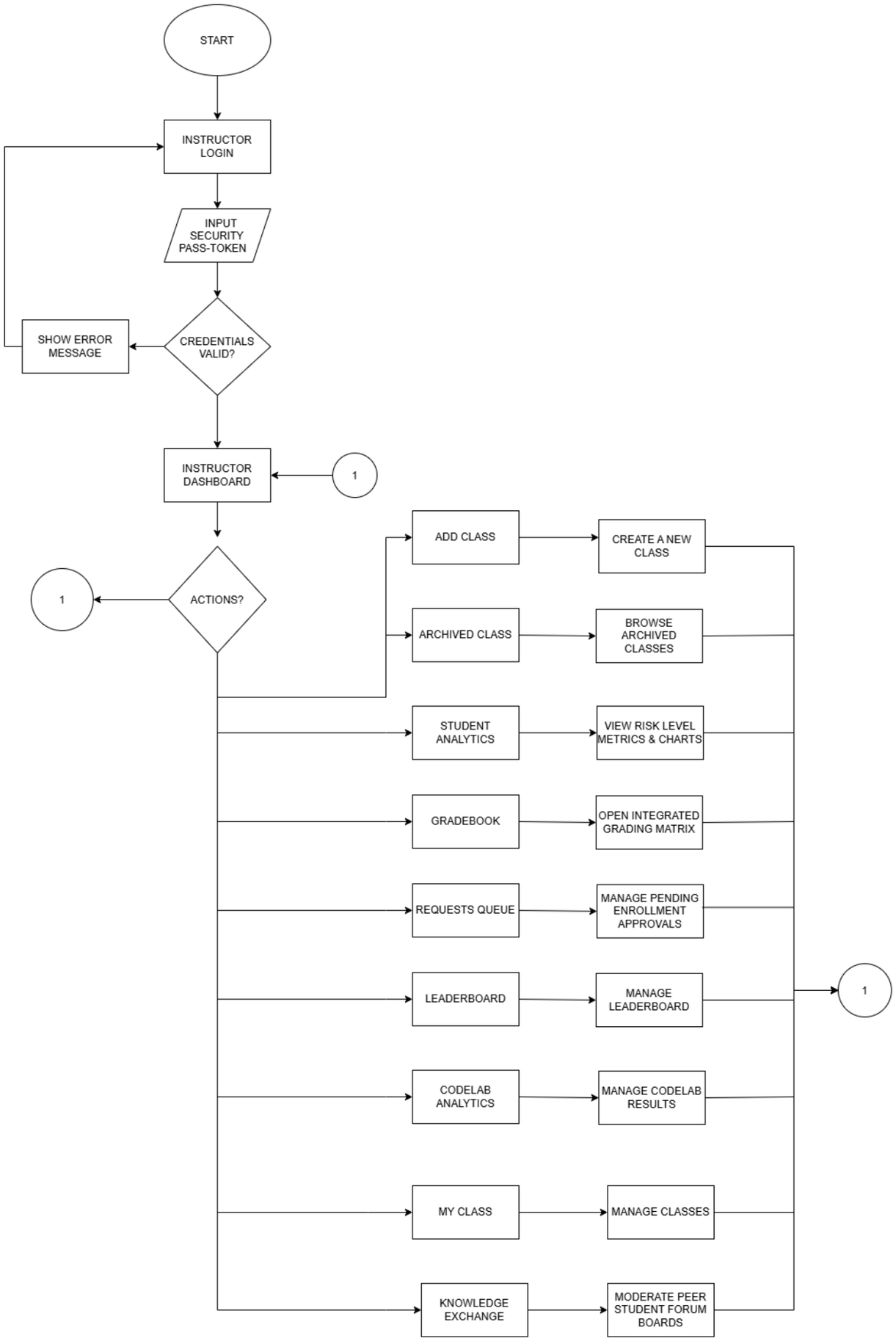
feature offers an environment in which students can carry out programming-related activities within the LMS.

4. Data Storage

The LMS makes use of its connected database and backend services in order to process and manage the information needed by its various features; for example, it retrieves the courses in which students are enrolled, processes enrolments in courses, displays the quizzes and assignments that have been assigned, manages the flashcards and tasks created by students, retrieves the leaderboard rankings, handles the discussions in the Knowledge Exchange, and provides support for CodeLab activities.

The specific information that each feature deals with is determined by its function and the operations which the system supports. Access to the records and the actions available are governed by the student's authorized permissions.

Instructor's View:



INSTRUCTOR VIEW

The instructor can create classes through the Instructor View of the Learning Management System (LMS).

- The instructor can also look at classes that have already been archived. The instructor keeps an eye on student risk-level metrics. The instructor manages grades. The instructor approves enrollment requests. The instructor handles leaderboards. The instructor looks at results from CodeLab. The instructor manages the classes. The instructor moderates discussions in Knowledge Exchange.

1. Authentication and Login

- The process starts when the instructor opens the Instructor Login page.
- The instructor then types login credentials and a security pass-token.
- The system receives these details for validation.

Login Validation

The system checks whether the credentials are correct.

If the credentials are invalid

- The system shows an error message.
- The instructor must go back to the Instructor Login page to try again.

If the credentials are valid

- The system grants access.
- The instructor then reaches the Instructor Dashboard.

The instructor can now use the dashboard.

2. Instructor Dashboard

After the instructor is logged in the instructor goes to the Instructor Dashboard, The Instructor Dashboard is the hub for all instructor functions. The instructor chooses an action to access the desired module.

The feature groups are shown in the flowchart.

A. Add Class

To create a class the instructor selects '**Add Class.**

This option starts the class setup process, inside the LMS.

B. Archived Class

To view archived classes the instructor selects '**Archived Class.**

This option lets the instructor look at classes that have been archived.

C. Student Analytics

The instructor selects **Student Analytics** to view risk level metrics and charts. Student Analytics shows student risk-level data and visual charts. Student Analytics helps instructors check the indicators of student performance.

D. Gradebook

The instructor selects **Gradebook** to open the Integrated Grading Matrix. Gradebook gives access to the grading interface. Gradebook lets instructors manage student grades using the system's functions.

E. Requests Queue

The instructor selects **Requests Queue** to access Manage Pending Enrollment Approvals. Requests Queue lets instructors review and handle enrollment requests that are still waiting for approval. Requests Queue lets instructors use the approval functions to make decisions.

F. Leaderboard

The instructor selects **Leaderboard** to access Manage Leaderboard. Leaderboard provides functions for managing the leaderboard inside the LMS. Leaderboard lets instructors review leaderboard data as needed.

G. CodeLab Analytics

The instructor selects CodeLab Analytics to access Manage CodeLab Results. **CodeLab Analytics** allows instructors to review and manage programming results from CodeLab. CodeLab Analytics lets instructors use the system's supported functions to track and manage student progress.

H. My Class

The instructor clicks on **My Class** to go to the Manage Classes section. This option allows the instructor to use tools for managing the classes they're responsible for.

I. Knowledge Exchange

The instructor clicks on **Knowledge Exchange** to get to the Moderate Peer Student Forum Boards area. This feature lets the instructor oversee student forum discussions and related content. The instructor can do this based on their assigned permissions.

3. Data Storage and Retrieval

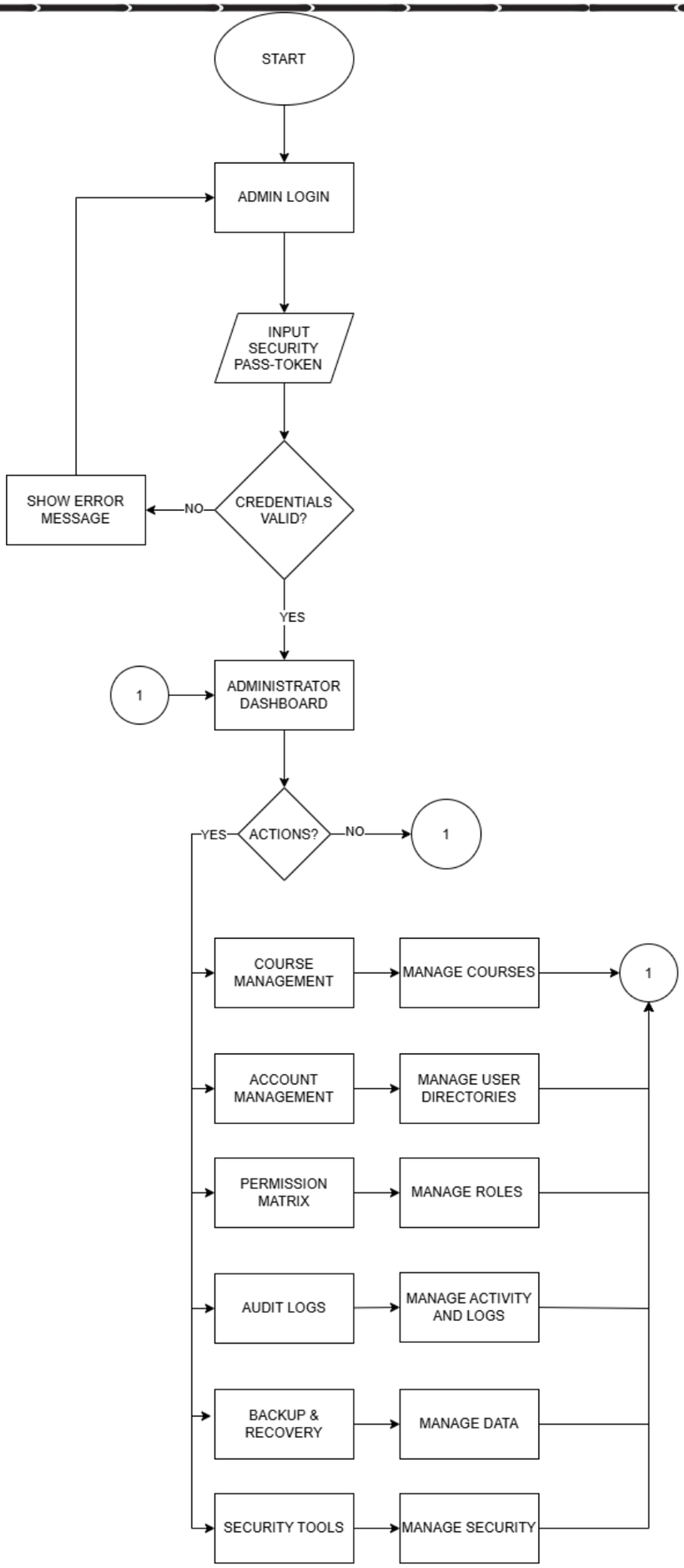
The Learning Management System uses its backend services and database to handle information needed by the instructor modules. Depending on the features used these operations may include saving and getting class details accessing classes pulling student data for analysis handling grade records managing enrollment requests keeping track of leaderboard data accessing CodeLab results organizing class settings and processing moderation actions in Knowledge Exchange.

Any changes to records or other system data must go through the proper backend services. These changes must follow the system's data management rules and the instructor's allowed permissions.

4. Completion of Instructor Actions

After using a feature and finishing the tasks the instructor can move to other functions using the Instructor Dashboard. The flowchart connects the dashboard to the rest of the feature paths using connector 1. It does not show a logout step or the full return route, from every module.

Admin's View:


START
ADMIN LOGIN
INPUT SECURITY PASS-TOKEN
CREDENTIALS VALID?
NO
SHOW ERROR MESSAGE
YES
1
ADMINISTRATOR DASHBOARD
ACTIONS?
YES
NO
1
COURSE MANAGEMENT
MANAGE COURSES
1
ACCOUNT MANAGEMENT
MANAGE USER DIRECTORIES
PERMISSION MATRIX
MANAGE ROLES
AUDIT LOGS
MANAGE ACTIVITY AND LOGS
BACKUP & RECOVERY
MANAGE DATA
SECURITY TOOLS
MANAGE SECURITY

ADMIN VIEW

The Administrator View in the Learning Management System (LMS) gives authorised administrators access to the various administrative functions needed for managing courses, user accounts, permissions, activity logs, data backup and recovery, security tools, and Knowledge Exchange. The process starts with the administrator being authenticated and then proceeds to the Administrator Dashboard, at which point the different administrative functions can be chosen.

1. Authentication and Login

The process starts when the administrator goes to the Admin Login page and enters the necessary login credentials together with the security pass-token, which are then sent to the system for validation.

Login Validation

The system checks whether the credentials submitted and the security pass-token are valid.

If the credentials are invalid:

- An error message is displayed by the system.
- The administrator is taken to the Admin Login page so that they can try once more.

If the credentials are valid:

- The system provides the administrator with access.
- The administrator is taken to the Administrator Dashboard.

2. Administrator Dashboard

Once the authentication process has been successfully completed, the administrator is brought to the Administrator Dashboard, which is the principal way of accessing the various administrative functions. The administrator then chooses an action in order to gain access to the relevant management module.

The flowchart shows the seven administrative feature groups.

A. Course Management

To use the Manage Courses function, the administrator must choose **Course Management**. The module enables the administrator to carry out the administrative management of the courses available in the LMS based on the permissions assigned to them.

B. Account Management

To access the Manage User Directories function, the administrator has to choose **Account Management**. The module offers the possibility of managing user directories in order to organize and keep the account information in the system.

C. Permission Matrix

To be able to use the Manage Roles function, the administrator must choose the **Permission Matrix.** This module offers the method of managing user roles together with the permissions connected to them, thereby allowing control over the functions to which authorised users are permitted to have access.

D. Audit Logs

In order to select the option called Manage Activity and Logs, the administrator must choose Audit Logs. This module enables the review and management of the system activities which have been recorded for the purpose of administrative monitoring and accountability.

E. Backup and Recovery

To be able to use the Manage Data function, the administrator must choose Backup & Recovery. The module contains the administrative data-management functions concerning backup and recovery, on the understanding that the functions implemented in the system allow this.

F. Security Tools

To access Manage Security the administrator chooses Security Tools. This module gives access to the system's various security-management functions which are used for maintaining administrative control and for protecting the system's resources.

G. Knowledge Exchange

To access the Manage Knowledge Exchange option, the administrator chooses Knowledge Exchange. The module enables the administrator to carry out the management of the Knowledge Exchange area using the administrative functions offered by the LMS.

3. Data management and system interaction

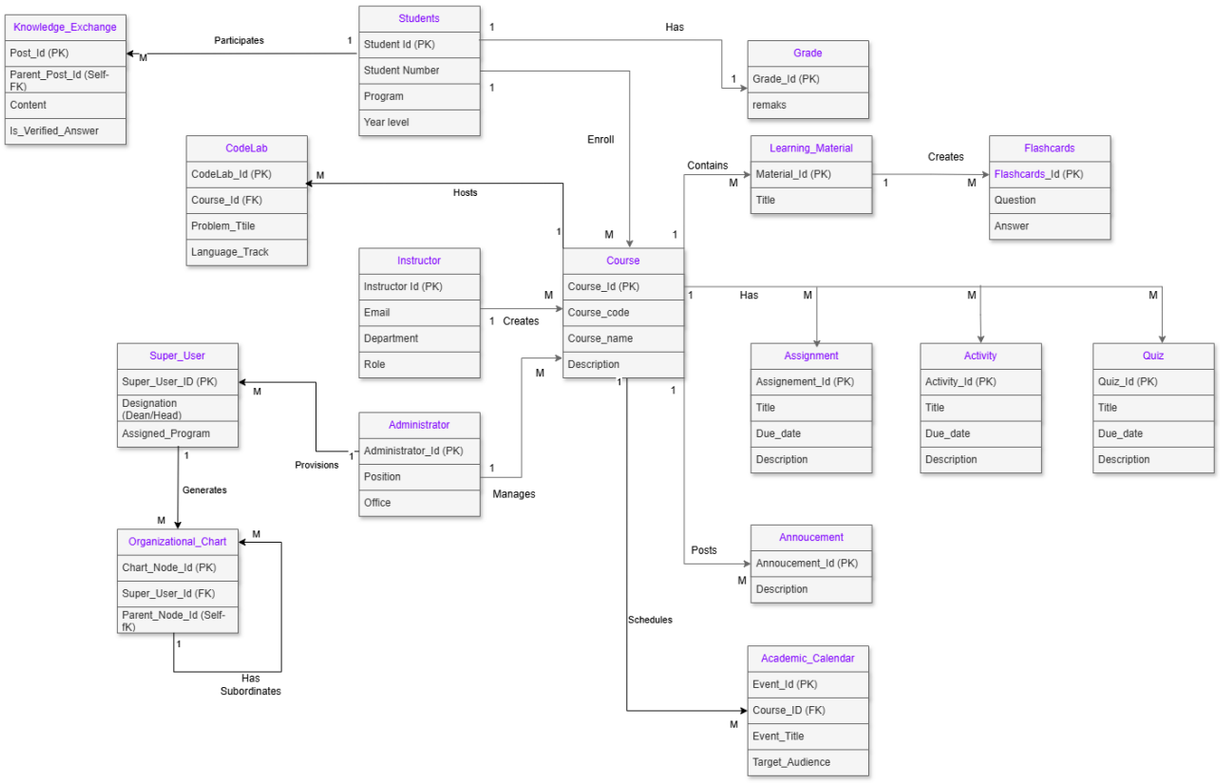
The Administrator View makes use of the LMS backend and database in order to handle administrative requests and keep track of system information. The specific operations involved will depend on the module that has been selected and the functions it provides, and these may include retrieving course records, accessing user directories, managing roles and permissions, reviewing activity logs, carrying out backup and recovery operations, managing security settings, and looking after Knowledge Exchange.

The administrative operations have to be carried out in accordance with the administrator's authorised permissions and any alterations to the system records must be validated and stored via the relevant backend services.

4. The carrying out of administrative actions

Once the administrator has selected and accessed a management module, the operations which are supported by that module can be carried out. The flowchart demonstrates the seven feature branches reconnecting via connector 1, thus showing that the administrative flow goes back to the shared continuation point. The diagram also fails to indicate a separate logout or system-exit process.

3.9 Entity Relationship Diagram



3.10 Data Flow Diagram